WAYAMBA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (Joint Major/ Special) Degree Examination
Level 3 Semester I– (Academic Year 2020/2021) June/July 2023

Department of Computing and Information Systems

CMIS 3142 – Computational Methods

No. of Pages: Three (03) *No. of Questions: Six (06)* *Time: Two (02) hours*

Answer four Questions only

1.

a. Convert the following decimal numbers into IEEE 754 single precision format.
   i. 654
   ii. -456
   iii. 512
   iv. -128 — 8 marks

b. Describe the difference between the IEEE 754 single precision and IEEE 754 double precision. — 4 marks

c. Write a Java program to convert a given decimal number into IEEE 754 double precision notation. — 8 marks

d. Write a Java method to convert IEEE 754 double precision notation into IEEE 754 single precision notation. — 5 marks

2.

a. Derive $\frac{\partial^2 f}{\partial x \partial y}$ from Taylor series using central difference, forward difference and backward difference where f is a function of x and y. — 9 marks

b. Comment on the truncation error in the equations in part (a). — 3 marks

c. Construct a forward difference table for $y = f(x) = x^3 + 2x + 1$ for $x = 1,2,3,4,5$ — 6 marks

d. Write an algorithm to construct the above table for $x = 1,2, \ldots ,25$. — 7 marks

3.

a. Describe forward error and backward error with suitable examples. 5 marks

b. Describe the impact of the rounding off error in the calculation of grades (result) to each course module and final GPA in the faculty of applied sciences. 8 marks

c. Propose a suitable method to minimize the loss in the calculation of grades of each course module and final GPA to students in the Faculty of Applied Sciences. 4 marks

d. Consider the following built-in methods in Java:

i. `Math.floor(a)` - rounds off a specified double value downward and returns it. (i.e. `Math_floor(3.30)` retirns 3.0)

ii. `Math.ceil(a)` - rounds off a specified double value upward and returns it. (i.e. `Math.ceil(3.30)` returns 4.00)

Write separate Java methods that serve the same purpose as each of the above two built-in methods. 8 marks

4.

a. Write an efficient Java class for a symmetric matrix with the following methods:

i. `element(i,j)` – returns the element at i^{th} row and j^{th} column of the matrix.

ii. `Store_Element(i,j, E)` – stores the element "E" of ith row and j^{th} column of the Matrix.

iii. `Multiply(A)` – returns the multiplication of the particular object with the symmetric matrix A. Note that the matrix returned may not be symmetric. 12 marks

b. Discuss how you can enhance the above program to facilitate a sparse symmetric matrix.(Only a discussion is sufficient; enhanced program is not needed) 5 marks

c. Let A be a row vector and B be a column vector. Write a separate Java method to perform each of the following calculations:

i. A*B

ii. B*A 8 marks

5.

a) Develop an algorithm to solve a given matrix equation $AX = B$ using the Cholesky decomposition method where A is a symmetric matrix, and X and B are column vectors. 9 marks

b) Using the algorithm you developed in part (a), solve the matrix equation given in part (a) where

$$A = \begin{bmatrix} 16 & 4 & 4 & -4 \\ 4 & 10 & 4 & 2 \\ 4 & 4 & 6 & -2 \\ -4 & 2 & -2 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 32 \\ 26 \\ 20 \\ -6 \end{bmatrix}$$ 10 marks

c) What are the limitations in the Cholesky method? 6 marks

6.

a. Describe the Gaussian elimination method to solve a given system of liner equations. 8 marks

b. A contractor categorizes the workers his employees as follows:

No	Group	Gender	Age range
1	Teen	Male and female	13 – 19
2	Man	Male	20 – 55
3	Woman	Female	20 – 50
4	Old	Male	Greater than 55

The contractor's Workers Grouping Quotation to complete a particular job is as follows:

8 teens, 2 men, 3 women, and 2 old will cost 189 $ per day.
3 teens, 6 men, 2 women, and 1 old will cost 192 $ per day.
1 teen, 2 men, 9 women, and 4 old will cost 233 $ per day.
2 teens, 3 men, 4 women, and 7 old will cost 224 $ per day.

Workers' daily wages per category is not mentioned in the quotation.

i. Convert the above real world scenario into a system of linear equations to find the daily wages of each category. 8 marks

ii. Solve the problem using the Gaussian elimination method. 9 marks